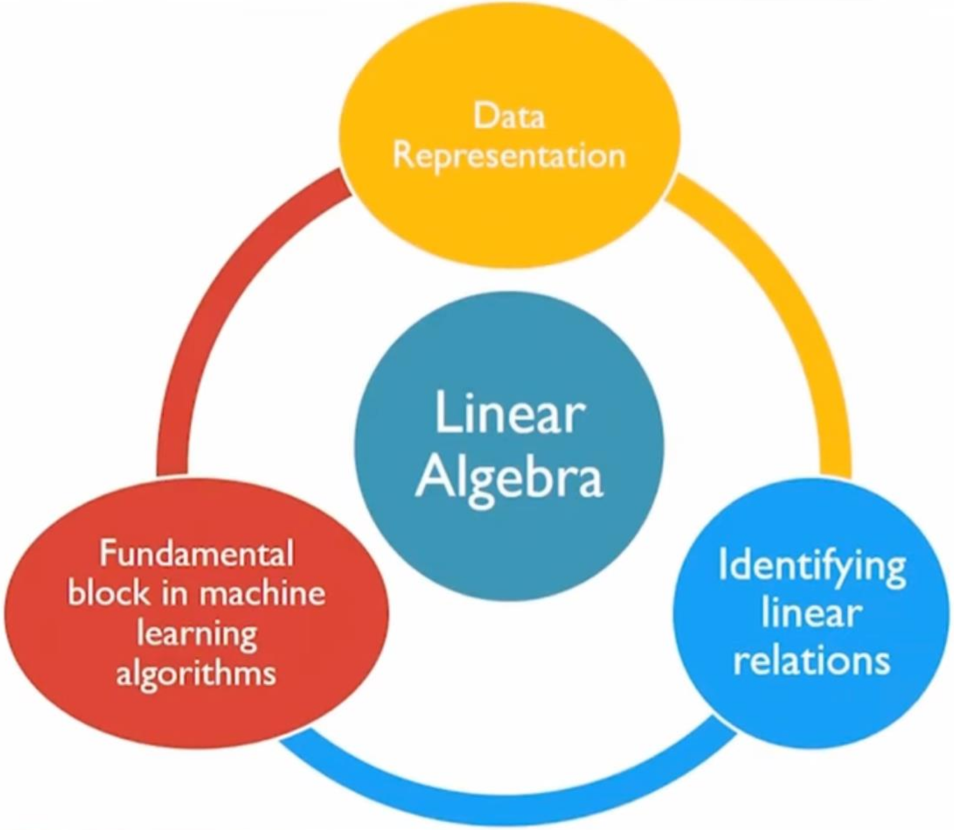
Overview


Data
Representation
Linear
Algebra
Fundamental
block in machine
learning
algorithms
Identifying
linear
relations

Matrix theory and linear algebra

Matrix Theory and Linear Algebra

- Matrices can be used to represent samples with multiple attributes in a compact form
- Matrices can also be used to represent linear equations in a compact and simple fashion
- Linear algebra provides tools to understand and manipulate matrices to derive useful knowledge from data

LINEAR ALGEBRA AND MATRICES

Matrices for data science: Data representation

- Usually matrices are used to store and represent the data on machines
- Matrix is a very natural approach for organizing data
- In general, data is organized in the following fashion
    - Rows represent samples
    - Columns represent the values of the variables (or attributes)
    - It is also possible to use rows for variables and columns for samples
    - However, we will stick to rows as samples and columns as variables in all of the material that will be presented

Data representation: Examples

- A real life example
  - Consider a reactor which needs to be controlled using multiple attributes from various sensors like Pressure (Pa), Temperature (K), Density (gm/m^3) etc.
  - Independently, the sensors have generated 1,000 data points
  - This complete set of information is contained in

$$
\begin{array}{c c}
 & \begin{array}{ccc} P & T & \rho \end{array} \\
\begin{array}{c} 1 \\ \vdots \\ 1000 \end{array} & \begin{bmatrix} 300 & 300 & 1000 \\ \vdots & \vdots & \vdots \\ 500 & 1000 & 5000 \end{bmatrix}
\end{array}
$$

Data representation: Examples

- Example 2:

$$X = [1,2,3]^T$$
$$Y = [2,4,6]^T$$

- X and Y are vectors pertaining to some attributes
- We define the A matrix using a column bind of X and Y thus representing data in a matrix format (the code for the same is attached)

R Code

```
x=c(1,2,3)
y=c(2,4,6)
A=cbind(x,y)
print(A)
```

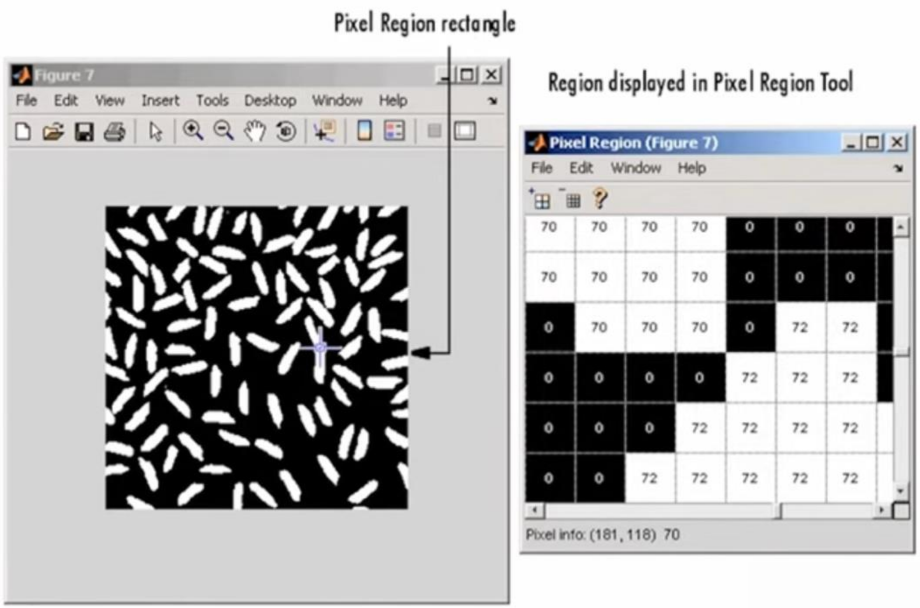
$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$$

Output

```
> print(A)
     x y
[1,] 1 2
[2,] 2 4
[3,] 3 6
```

Data representation: Examples

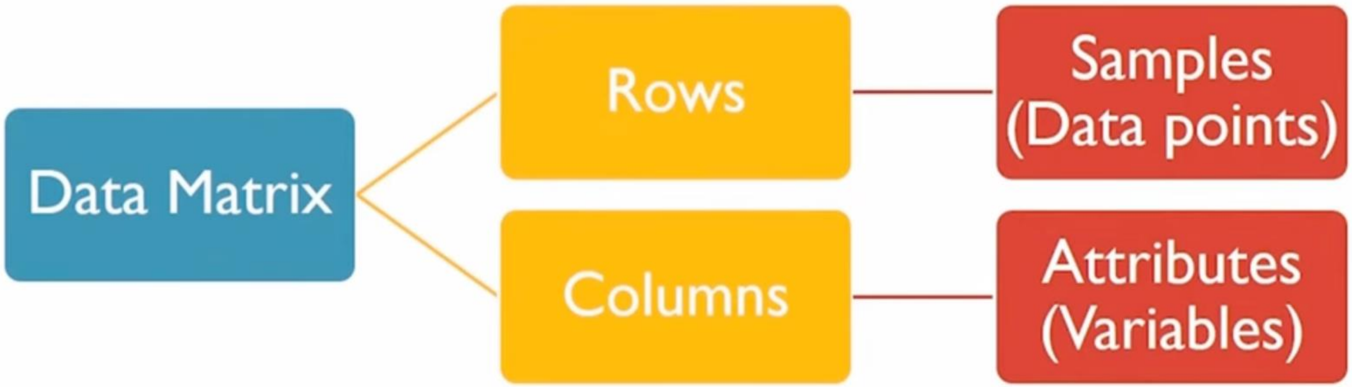
- The simplicity in representation will become apparent when the image below is considered



Data representation: Examples

- Storing
  - The image is stored in the machine as a large matrix of pixel values across the image.
  - Thus, storing the pixel value matrix is equivalent to storing the image for the machine
- Identification
  - Several machine learning algorithms are deployed in order to "teach" the machine how to identify a particular image.
  - Linear algebra and matrix operations are at the heart of these machine learning algorithms.

Data as matrix: Summary



IDENTIFICATION OF INDEPENDENT ATTRIBUTES

Further analysis

- Now that we can represent the data into a matrix format, we ask the following questions
  - Are all the attributes in the data matrix relevant/ important?
  - Is there any method which can identify if some attributes are related to the other attributes?
  - If yes, how do we identify the linear relationship?
  - Can we use this to reduce the size of the data matrix?

Identification of independent attributes: Example

- Consider the ideal reactor example with multiple (say, 4) attributes like Pressure, Temperature, Density, Viscosity, etc. with 500 samples.
- Thus we have a $500\ X\ 4$ matrix such that
$$A = [P\ T\ D\ \eta]$$
- P, T, D and η are vectors of 500 samples from the pressure, temperature, density and viscosity sensors.
- How does one identify the number of independent attributes?

Identification of independent attributes: Example

- Domain knowledge

$$D \sim f(P, T)$$

- Thus, in some sense **D** is a function of **P** and **T**
- Implying that at least one attribute is dependent on the others
- This variable can be calculated as a linear combination of the other variables
- The physics of the problem helps us identify the relationship in the data matrix
- We now ask if the data itself will help us identify these relationships

Number of independent attributes: Rank of a matrix

- Let us assume that we have many more samples than attributes for now
- Is there any approach which can be used to identify the number of linear relationships between the attributes purely using data?
- This is addressed by the concept of the **rank** of the matrix.
- **Rank** of a matrix refers to the number of linearly independent rows or columns of the matrix
- The rank of a matrix can be found using the rank command: rank(A)

IDENTIFICATION OF LINEAR RELATIONSHIPS AMONG ATTRIBUTES

Null space for data science

- The null space of a matrix **A** consists of all vectors $\boldsymbol{\beta}$ such that $\boldsymbol{A\beta} = \mathbf{0}$ and $\boldsymbol{\beta} \neq \mathbf{0}$
- Nullity of a matrix is the number of vectors in the null space of the given matrix
- The size of the null space of a matrix provides us with the number of linear relations among the attributes
- And the null space vectors $\boldsymbol{\beta}$ are useful to identify these linear relationships

SOLVING MATRIX EQUATIONS

Preliminaries

- We consider the following set of equations

$$\boldsymbol{Ax} = \boldsymbol{b}$$

$$\boldsymbol{A}(m \times n);\ \boldsymbol{x}(n \times 1);\ \boldsymbol{b}(m \times 1)$$

- Generalized linear equations can be represented in the above format.
- m and n are the number of equations and variables respectively.
- $\boldsymbol{b}$ is the general RHS commonly used

Categorization

$m = n$	• Number of equations and variables are the same • Easiest case to solve
$m > n$	• More equations than variables • Usually no solution
$m < n$	• Number of equations less than number of variables • Usually multiple solutions

We look into these cases independently

Full row and column rank: Concepts

- Consider a matrix data matrix $\boldsymbol{A}$ $(m \times n)$

Full Row Rank

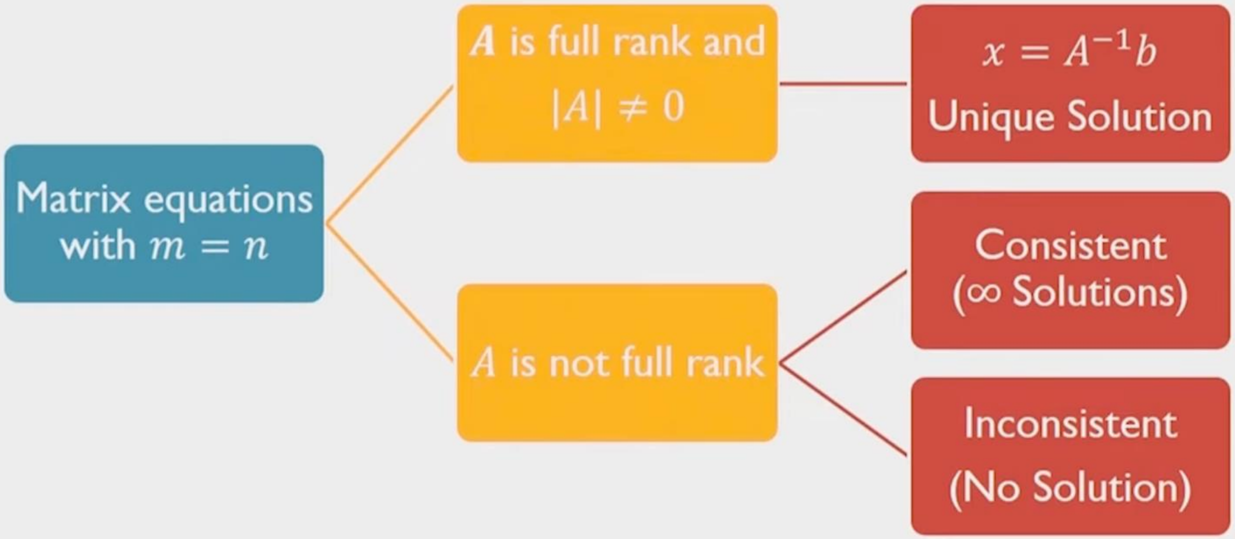
- When all the rows of the matrix are linearly independent
- Data sampling does not present a linear relationship – samples are independent

Full Column Rank

- When all the columns of the matrix are linearly independent
- Attributes are linearly independent

Row rank = Column rank

Case 1: $m = n$


Matrix equations with $m = n$
A is full rank and $|A| \neq 0$
$x = A^{-1}b$
Unique Solution
A is not full rank
Consistent
(∞ Solutions)
Inconsistent
(No Solution)

Case 1: Example 1.1

$$\begin{bmatrix}1 & 3\\2 & 4\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}7\\10\end{bmatrix}$$

R Code

```
A=matrix(c(1,2,3,4),ncol=2, byrow=F)
b=c(7,10)
x=solve(A)%*%b
```

$$|\boldsymbol{A}| \neq 0$$

$$rank(\boldsymbol{A}) = 2 = no.\,of\ columns$$

Console output

```
> x
     [,1]
[1,]    1
[2,]    2
```

- This implies that A is full rank

$$\begin{bmatrix}x_1\\x_2\end{bmatrix}=\begin{bmatrix}1 & 3\\2 & 4\end{bmatrix}^{-1}\begin{bmatrix}7\\10\end{bmatrix}=\begin{bmatrix}-2 & 1.5\\1 & -0.5\end{bmatrix}\begin{bmatrix}7\\10\end{bmatrix}=\begin{bmatrix}1\\2\end{bmatrix}$$

- Thus, the solution for the given example is $(x_1, x_2) = (1,2)$

Case 1: Example 1.2

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$|\boldsymbol{A}| = 0; \; rank(\boldsymbol{A}) = 1; nullity = 1$

- Checking consistency

$$\begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$Row(2) = 2Row(1)$$

- The equations are consistent with only one linearly independent equation
- The solution set for (x_1, x_2) is infinite because we have only one linearly independent equation and 2 variables

Case 1: Example 1.3

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

$$|\boldsymbol{A}| = 0$$
$$rank(\boldsymbol{A}) = 1$$
$$nullity = 1$$

- Checking consistency

$$\begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$
$$2Row(1) = 2x_1 + 4x_2 = 10 \neq 9$$

- Thus the equations are inconsistent
- One cannot find a solution to (x_1, x_2)

Case 2: $m > n$

- This is the case of not enough variables or attributes
- Since the number of equations is greater than the number of variables, in general, not all equations can be satisfied
- Hence it is sometimes termed as a no-solution case
- However, we can identify an appropriate solution by viewing this case from an optimization perspective

Case 2: An optimization perspective

- Instead of identifying a solution to $Ax - b = 0$, one can identify an x such that $(Ax - b)$ is minimized
- Notice that $(Ax - b)$ is a vector
- There will be as many error terms as the number of equations
- Denote $(Ax - b) = e(mx1)$; there are m errors $e_i, i = 1{:}m$
- One could minimize all the errors collectively by minimizing $\sum_{i=1}^{m} e_i^2$
- This is the same as minimizing $(Ax - b)^T(Ax - b)$

Case 2: An optimization perspective

- This optimization problem is

$$\min[(Ax-b)^T(Ax-b)]$$
$$=\min[(b^T-x^TA^T)\,(Ax-b)]$$
$$=min[(x^TA^TAx-2b^TAx+b^Tb)=f(x)]$$

- We observe that the optimization problem is a function of x
- Solving the optimization problem will result in a solution for x
- The solution to this optimization problem is obtained by differentiating $f(x)$ with respect to x and setting the differential to zero

$$\nabla f(x)=0$$

Case 2: An optimization perspective

- Differentiating $f(x)$ and setting the differential to zero results in

$$2(A^T A)x - 2A^T b = 0$$
$$(A^T A)x = A^T b$$

- Assuming that all the columns are linearly independent

$$\boldsymbol{x} = (\boldsymbol{A}^T \boldsymbol{A})^{-1} \boldsymbol{A}^T \boldsymbol{b}$$

Case 2: Example - I

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

- $m = 3, n = 2$
- Using the optimization concept,

$$\boldsymbol{x} = (\boldsymbol{A}^T \boldsymbol{A})^{-1} \boldsymbol{A}^T \boldsymbol{b}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

Case 2: Example continued

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.2 & -0.6 \\ -0.6 & 2.8 \end{bmatrix} \begin{bmatrix} 15 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

- Thus, the solution for the given example is $(x_1, x_2) = (0,5)$
- Substituting in the equation shows

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} \neq \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

Case 2: Example

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

- $m = 3, n = 2$
- Using the optimization concept,

$$\boldsymbol{x} = (\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T\boldsymbol{b}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

Case 2: Example continued

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.2 & -0.6 \\ -0.6 & 2.8 \end{bmatrix} \begin{bmatrix} 20 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- Thus, the solution for the given example is $(x_1, x_2) = (1,2)$
- Substituting in the equation shows

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix} \doteq \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

R Code

```
A=matrix(c(1,2,3,0,0,1),ncol=2, byrow=F)
b=matrix(c(1,2,5),ncol=2, byrow=F)
x=inv(t(A)%*%A)%*%t(A)%*%b
x
```

Console output

```
> x=inv(t(A)%*%A)%*%t(A)%*%b
> x
     [,1]
[1,]    1
[2,]    2
>
```

Case 3: $m < n$

- This case addresses the problem of more attributes or variables than equations
- Since the number of attributes is greater than the number of equations, one can obtain multiple solutions for the attributes
- This is termed as an infinite-solution case
- How does one choose a single solution from the set of infinite possible solutions?

Case 3: An optimization perspective

- Pose the following optimization problem

$$\min\left(\frac{1}{2}x^T x\right) \text{ s.t. } Ax = b$$

- Define a Lagrangian function $f(x,\lambda)$

$$\min\left[f(x,\lambda) = \frac{1}{2}x^T x + \lambda^T(Ax - b)\right]$$

- Differentiating the Lagrangian with respect to x, and setting to zero

$$x + A^T\lambda = 0$$

Case 3: An optimization perspective

$$x = -A^T\lambda$$

Pre-multiplying by A

$$Ax = b = -AA^T\lambda$$

Thus we obtain $\lambda = -(AA^T)^{-1}b$ assuming that all the rows are linearly independent

$$\boldsymbol{x} = -\boldsymbol{A}^T\boldsymbol{\lambda} = \boldsymbol{A}^T\left(\boldsymbol{A}\boldsymbol{A}^T\right)^{-1}\boldsymbol{b}$$

Case 3: Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- $m = 2, n = 3$
- Using the optimization concept,

$$\boldsymbol{x} = \boldsymbol{A}^{\boldsymbol{T}} \left(\boldsymbol{A}\boldsymbol{A}^{\boldsymbol{T}} \right)^{-\mathbf{1}} \boldsymbol{b}$$

$$x = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Case 3: Example

$$x = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 14 & 3 \\ 3 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} -0.2 \\ 1.6 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -0.2 \\ -0.4 \\ 1 \end{bmatrix}$$

- The solution for the given example is $(x_1, x_2, x_3) = (-0.2, -0.4, 1)$

R Code

```
A=matrix(c(1,0,2,0,3,1),ncol=3)
b=c(2,1)
library(MASS)
x = t(A)%*%inv(A%*%t(A)) %*%b
x
```

Console output

```
A=matrix(c(1,0,2,0,3,1),ncol=3, byrow=F)
b=c(2,1)
x = t(A)%*%inv(A%*%t(A)) %*%b
x
```

Case 3: Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- The solution for the given example is $(x_1, x_2, x_3) = (-0.2, -0.4, 1)$
- Verify this is a solution that satisfies the original equation
- This also turns out to be minimum norm solution

Generalization

- The described cases cover all the scenarios one might encounter while solving linear equations
- Is there any form in which the results obtained for cases 1, 2 and 3 can be generalized ?
- The concept we used to generalize the solutions is called as Moore-Penrose pseudo-inverse of a matrix
- The pseudo inverse is used as follows

$$Ax = b$$

The solution becomes

$$x = A^{+}b$$

- Singular Value Decomposition can be used to calculate the pseudo inverse or the generalized inverse (A^{+})

Vector Spaces

what are vector spaces & subspaces?

what do we do with vectors? → We add them or multiply them with a scalar.

so for vector spaces, we shall be able to add vectors or multiply them with numbers, & some decent rules shall be satisfied!

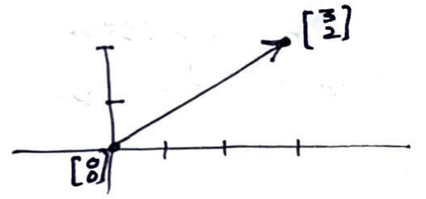
- closure under addition.
- $U+V = V+U$
- $(U+V)+W = U+(V+W)$
- $U+0 = U$ Additive identity
- $U+(-U) = 0$

Example:

Space means a bunch of vectors but not any bunch of vectors.

$\alpha V_1 + \beta V_2$ in the same space

$\mathbb{R}^2$: All 2D real vectors $\begin{bmatrix}3\\2\end{bmatrix}, \begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}\pi\\e\end{bmatrix}$ . . .



& R^2 is the whole XY plane

however we say R^2 is a vector space because all such vectors live there.

e.g we cannot take out $\begin{bmatrix}0\\0\end{bmatrix}$, bcz we should have the freedom of multiplying with 0 to reach a $\begin{bmatrix}0\\0\end{bmatrix}$ vector. Similarly $\begin{bmatrix}3\\2\end{bmatrix}+\left(-\begin{bmatrix}3\\2\end{bmatrix}\right)$ can not be done without the origin.

every vector space got to have $\begin{bmatrix}0\\0\end{bmatrix}$ vector in it.

Similarly $\mathbb{R}^3$: vectors with 3 real components

$\begin{bmatrix}3\\2\\0\end{bmatrix}$

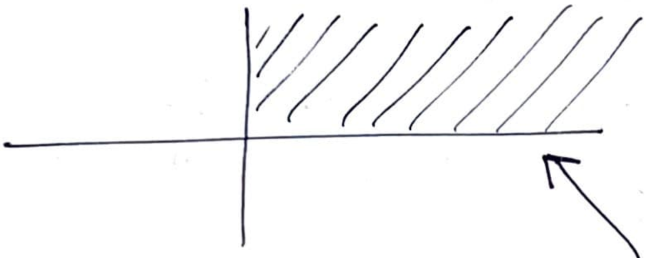
So is

R^n : all column vectors with n real components

So the question is: can we do such additions & do we stay in the space?

A case where you can't!

not a vector space



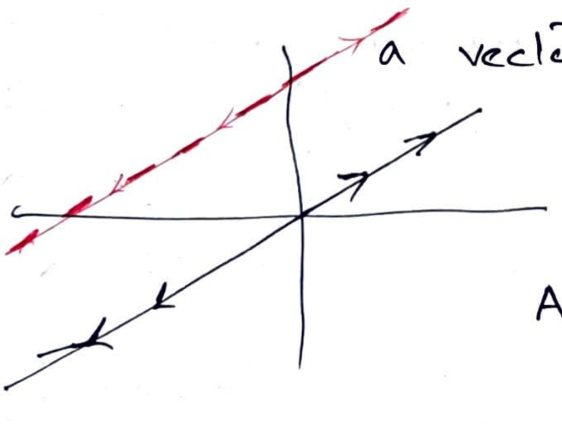
← is not a vector space bcz it is not closed under multiplication.

bcz scalar can be any value...
although sum of two vectors will certainly reside in this quarter space.

A vector space has to be closed under addition as well as multiplication.

* We just took part of R^2 & messed up (the part did not turn out to be a vector space)

* Think of a vector space that is part of R^2 & is still a vector space? (Subspace)



a vector space inside R^2
a subspace of R^2

A line in R^2 is a subspace.

But not just any line!

* adding two vectors on this line will still be on the line & multiplying with any scalar be still on the line.

Vector Spaces

→ Listing subspaces of R^2

1. All of R^2 ← Plane
2. Any line through $\begin{bmatrix}0\\0\end{bmatrix}$ ← Line
3. $\begin{bmatrix}0\\0\end{bmatrix}$ ← Zero Vector / A point

→ For R^3

1. all of R^3
2. Any plane through the origin
3. Line through the origin
4. Zero vector $\begin{bmatrix}0\\0\\0\end{bmatrix}$

The interesting real question is, where do these subspaces come... how do they come out of matrices?

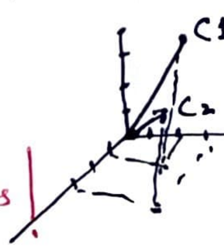
Take a matrix

$$A = \begin{bmatrix}1 & 3\\2 & 3\\4 & 1\end{bmatrix}$$

∴ columns of A are vectors in R^3

if I want to create a subspace from these two cols, can't just put these two columns in a subspace.
what else is needed?

* → All the linear combinations of these vectors will form a subspace, called the Column Space of A
$C(A)$



* How to understand $Ax=b$ in this new language of column spaces

how shall the subspace look like.

Rem: linear combination of C1 & C2

Imagine it
Visualize it

(Passing through origin) ⇒ Plane in R^3

* Subspaces of 3D Space.

$x_1 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ $x_2 = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}$

→ Find V_1 = subspace generated by x_1 → "Subspace" line in $\mathbb{R}^3$

V_2 = " " " x_2 → Line in $\mathbb{R}^3$

Describe $V_1 \cap V_2$ → a point in $\mathbb{R}^3$ also a subspace $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

→ Find V_3 = subspace generated by $\{x_1, x_2\}$ → A plane in $\mathbb{R}^3$

Is $V_3 = V_1 \cup V_2$? NO $\because x_1 + x_2 \notin V_1 \cup V_2$ is not a subspace

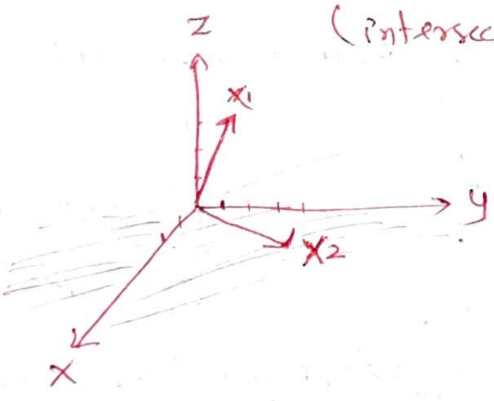
Find a subspace S of V_3 s.t. $x_1 \notin S$ $x_2 \notin S$

Actually yes, it could be a line e.g. $x_1 + x_2$ pass thru 0.

→ What is $V_3 \cap \{$x-y plane : a subspace of $\mathbb{R}^3\}$.

A subspace a line in $\mathbb{R}^3$. along x_2. actually V_2.

(intersection of two planes).



⇓

intersection of two subspaces of $\mathbb{R}^3$ is again a subspace of $\mathbb{R}^3$.

Column Space & Null Space

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$$

- $C(A)$ is subspace of R^4
- whats in this space?
  ⇒ Linear combinations of ~~C(A)~~ indep. columns of A, & offcourse the NULL vector.

* How big is this space? Is this space the whole four dimentional space? NO Its a plane in R^4.

. if we start with 3 vectors we cant get the whole 4D space. ∴ two indep vectors/basis.

* Behind all this study we have a purpose!

* Does $Ax = b$ always have a solution for every R.H.S b ?

→ NO

which R.H.S b's are okay → Only for bunch of b's that belong to $C(A)$
∴ b = linear comb. of columns of A.

→ There can be lots of b's that are not on the plane span by columns of A, yet these b's are in R^4 & could form a subspace of R^4.

Which R.H.S allows me to solve $Ax = b$ *

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \text{ or } \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \text{ or } \begin{bmatrix} 4 \\ 6 \\ 8 \\ 10 \end{bmatrix} \text{ or even } \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Sol $\quad \bar{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \bar{x} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \bar{x} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$

Can we Throw away any columns of A & still have the same column space?

$\therefore$ C_3 is $C_1 + C_2$ so redundant

$\therefore$ C_1 is $\frac{1}{2} C_3$ so C_3 could also be kept.

So $C(A)$ is a 2D subspace of R^4

NULL SPACE

A totally diff subspace. Does'nt contain R.H.S but x's, All x's that solve $Ax=0$

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$$

$\therefore$ the null space is a subspace of R^3.

So what is in the null space, while I want to solve these four equations

$$AX = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

always a trivial solu. we are not interested in.

seeking any non-zero in x_1, x_2, x_3 that satisfies the equations.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} c \\ c \\ -c \end{bmatrix}$$

family of solutions. (also accommodates $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$) forms a subspace. called Null space of A.

$N(A)$ is a subspace in R^3

* It is a line passing through origin & $\begin{bmatrix} c \\ c \\ -c \end{bmatrix}$.

~~Significance of NULL SPACE~~:

first lets go back & check, how do we ~~say~~/establish that the null space is a vector space?

so check: solutions to $Ax=0$ always give a subspace!

Basically if U & W $\in N(A)$, hence sols. to $Ax=0$ then their sum shall also belong to $N(A)$ & indeed a sol to $Ax=\bar{0}$. That means:

if $AU=\bar{0}$ & $Aw=\bar{0}$ then $A(U+W)=\bar{0}$

because linearity allows
$$AU + Aw = \bar{0}$$
$$\bar{0} + \bar{0} = \bar{0}$$

Also any multiple $\beta(U+w)$ also satisfies

$$A(\beta U + \beta w) = 0$$

$$\text{as } \beta . \underbrace{A(U+w)}_{0 \text{ already proved}} = 0$$

$\Rightarrow$ $N(A)$ is a subspace!

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whats the point of a vector space?

Two ways of forming a subspace: either you have freedom to choose x's & form the $C(A)$ by just storing $A \cdot X \to b \to C(A)$. or you are given a system of eq to solve & you find x's that solve $Ax=0$, forming a subspace. By the way does the sol's to

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

Do the solutions form a subspace?

$\boxed{\text{NO}}$ $\rightarrow$ form a plane that does not pass thru $\bar{0}$

Significance of NULL SPACE?

1- <u>uniqueness of Solutions</u>: if the null space contains more than just the zero vector, then there are infinitely many solutions to $Ax = b$.

2- <u>detecting redundancy</u>: null space shows combinations of input features i.e Domain (A) i.e where all xs lie, that have no effect on the output indicating redundant or dependent features in data — useful in machine learning for feature selection & dimentionality reduction.

The NULL SPACE is like walking in directions inside a space where your movements leave no trace — because the transformation "erases" them. (moving in the input space without creating any effect on the output)

Linear Independence, Spanning a Space, Basis & Dimension

Talking about a bunch of vectors:

- being independent
- spanning a space i.e being a basis
- dimension of the basis.

Background

Suppose A is $m \times n$ with $m < n$, then there are non-zero solutions to $Ax = 0$

Reason: more unknowns than equations, there will be free variables!!!

So what does it mean for bunch of vectors to be independent?

The vectors $x_1, x_2, \ldots x_n$ in a vector space are independent if no linear combination gives the zero vector. (except the zero combination)

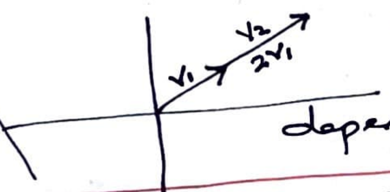
(simplest analogy two vectors gets subtracted to give 0 vector hence both were actually on the same line.
Higher concept: This mean if a line is a subspace it can atmost have only one independent vector.
This is also called dimension)

So any comb

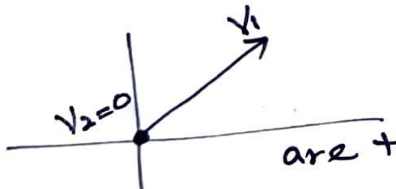
$$c_1x_1 + c_2x_2 + \ldots c_nx_n \neq 0$$

except if all c's $= 0$

so $2v_1 - v_2 = 0$



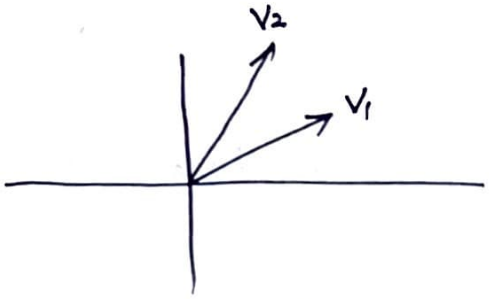
dependent



are they indep?

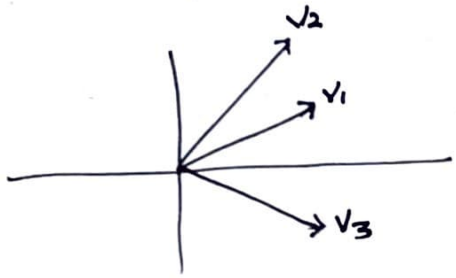
NO $v_2 = 0 \times v_1$

* if one of these vectors is the zero vector then I could always include that one & none of others & I would get the zero answer & it would show dependence!



They are certainly independent

BUT



In 2D space, I have 3 vectors they must be dependent Because some combination of V_1, V_2 & V_3 will give zero vector.

$$A = \begin{bmatrix} V_1 & V_2 & V_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 & 2 & -2 \\ 1 & 3 & -1 \end{bmatrix}}_{\text{example not taken carefully}} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

sol. to above system ensures there exist a null space, so the columns in A are dependent.

Again:

To determine some vectors are independent or not put them in a matrix & solve $Ax = 0$ & if the null space of A is the zero vector then the vectors substituted as columns of A should be independent.

They are dependent if $Ax = 0$ for some non-zero x in $N(A)$ implying there is some comb. of columns gives zero vector.

RANK $= n$ & $N(A) = \{0\}$ no free variables.

RANK $< n$ & yes free variables.

*In ML it helps again in finding independent features as each feature is represented as a vector!

Vector Spaces

* What does it mean for a bunch of vectors to span a space?

Vectors $v_1, v_2 \dots v_p$ SPAN a space means: The space consist of all combinations of those vectors.

- The columns of a matrix spans the column space.

But are these vectors/columns independent? Obviously we are interested in a set of vectors that spans a space & is independent.

A $\boxed{\text{Basis}}$ for a vector space is a sequence of vectors $v_1, v_2 \dots v_d$ with two properties

i) independent
ii) they span the space.

if you give me a Basis for a subspace, you have told me everything I need to know about the subspace!

Example: Basis for $\mathbb{R}^3$

$\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$ are these indep? Take their L.C & see if we can get zero vector? $\boxed{\text{Never}}$ only if all c's are zero.

→ That is not the only basis.

$\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}2\\2\\5\end{bmatrix}$. are these indep? Yes

. do they span $\mathbb{R}^3$? NO ∴ there are some vectors in $\mathbb{R}^3$ that can't be spanned by these what about selecting the third one?

$\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}2\\2\\5\end{bmatrix}, \& \begin{bmatrix} \\ \\ \end{bmatrix}$ (any vector that is NOT in the plane spanned by first two vectors).

$\begin{bmatrix}3\\3\\8\end{bmatrix}$

* What is the test/check if the given set is a basis!

* Put in a matrix & see if the matrix is invertible?

* There are infinite no. of basis. Any 3x3 invertible matrix is a basis for R^3.

But there is something common with all these Basis, that these Basis share:

→ They all shall have the same no. of vectors

that number is called the "Dimension of the space"

Example: $A = \begin{bmatrix}1&2&3&1\\1&1&2&1\\1&2&3&1\end{bmatrix}$

- Does the columns span the C(A)? Yes
- Are they ~~Basis~~ indep.? No they are NOT INDEPENDENT
- These is something in N(A) e.g some x that make Ax=0

$\begin{bmatrix}-1\\0\\0\\1\end{bmatrix}$ or $\begin{bmatrix}1\\0\\0\\-1\end{bmatrix}$ or $\begin{bmatrix}-1\\-1\\1\\0\end{bmatrix}$

* What could be the basis for the column space?

C1 & C2 so the RANK(A) = 2 = DIMENSION of C(A)

- Any other Basis for the C(A)?

C1 & C3 or C2 & C4 or $\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$ that is not there in A but span C(A).

* What is the dim of N(A)?

should be n - rank(A). or n - dim (C(A))

where n = no. of columns of A.

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